Lecture 9

- KF min mean square optimality
- Steady state KF
- Extended Kalman Filter (EKF)

Recall we saw L-G ss model:

(i) jointly Gaussian $(X_{0:t}, Y_{1:t})$

(ii) Markovianity

(iii) Cond. indep. measurements

Also Kalman filter is the recursive Bayesian estimator for the L-G model:

$$\boxed{p(x_t \mid y_{1:t}) = \mathcal{N}(\mu_{t|t}, \Sigma_{t|t})}$$

from KF

Q: In what sense is the KF optimal?

A: Mean squared error (MSE) sense

Minimum Mean Squared Error (MMSE) Estimation

consider a state estimator,

state: x is unknown, y known

$\hat{x}_{MMSE}(y)$ "as close as possible to x"

let's choose to model x as a R.V. X

suppose I know $p(x|y)$ relating known y to unknown x.

suppose I want $\hat{x}(y)$ to minimize:

$$\hat{x}_{MMSE}(y) = \underset{\hat{x}(y)}{\operatorname{argmin}} \, E\left[\|\hat{x}(y) - x\|^2 \mid Y = y\right]$$

"min" "mean" squared error := MMSE

notice:

$$\min_{\hat{x}(y)} \int_x \|\hat{x}(y) - x\|^2 p(x|y)\, dx$$

↳ diff w.r.t. $\hat{x}$

$$\int_x 2(\hat{x} - x)\, p(x|y)\, dx = 0$$

$$2\hat{x} \underbrace{\int p(x|y)\, dx}_{1} - 2\int_x x\, p(x|y)\, dx = 0$$

$$E[X \mid Y = y]$$

we find:

$$\boxed{\hat{x}_{MMSE}(y) = E[X \mid Y = y]}$$

- true for any Bayesian posterior
- specifically true for KF

we have:

$$p(x_t \mid y_{1:t}) = \mathcal{N}(\mu_{t|t}, \Sigma_{t|t})$$

$$= E[X_t \mid Y_{1:t}]$$

so $$\boxed{\hat{x}_{MMSE\,t|t} = E[X_t \mid Y_{1:t}] = \mu_{t|t}}$$

from KF

Steady State Kalman Filter

Suppose: (A, B, C, Q, R) time-invariant

then the covariance KF eqns are:

predict: $\Sigma_{t|t} = A\Sigma_{t|t} A^T + Q$ (i)

update: $\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{t|t-1} C^T (C\Sigma_{t|t-1} C^T + R)^{-1} C\Sigma_{t|t-1}$ (ii)

sub (i) into (ii) to get single cov. update + predict:

$$\Sigma_{t+1|t} = A\Sigma_{t|t-1} A^T - A\Sigma_{t|t-1} C^T (C\Sigma_{t|t-1} C^T + R)^{-1} C\Sigma_{t|t-1} A^T + Q$$

claim: under some conditions (later) $\Sigma_{t+1|t}$ converges to a SS value:

$$\lim_{t\to\infty} \Sigma_{t+1|t} = \Sigma_{ss}$$

we get a algebraic eqn for Σ_{ss}:

$$\boxed{\Sigma_{ss} = A\Sigma_{ss} A^T + A\Sigma_{ss} C^T (C\Sigma_{ss} C^T + R)^{-1} C\Sigma_{ss} A^T + Q}$$

Discrete Time Algebraic Riccati Eqn (DARE)

Then we can compute steady state Kalman gain:

$$\boxed{K_{ss} = \Sigma_{ss} C^T (C\Sigma_{ss} C^T + R)^{-1}}$$

we get ss Kalman Filter:

$$\boxed{\begin{aligned} &\text{predict: } \mu_{t|t-1} = A\mu_{t-1|t-1} + Bu_{t-1} \\ &\text{update: } \mu_{t|t} = \mu_{t|t-1} - K_{ss}(y_t - C\mu_{t|t-1}) \end{aligned}}$$

We pre-compute K_{ss} before deployment.

Notice: This is a Luenberger Observer (from state space control)

Q: under what conditions does Σ_{ss} exist?

(i) (A, B, C, Q, R) time-invariant

(ii) (A, C) observable, so

$$\mathcal{O} := \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$ observability matrix is full rank $\text{rank}(\mathcal{O}) = n$.

(iii) $(A, Q^{1/2})$ controllable, so

$\mathcal{C} := [Q^{1/2} \; AQ^{1/2} \; \dots \; A^{n-1}Q^{1/2}]$ is full rank, $\text{rank}(\mathcal{C}) = n$.

these are sufficient for a unique solution to DARE to find Σ_{ss}.

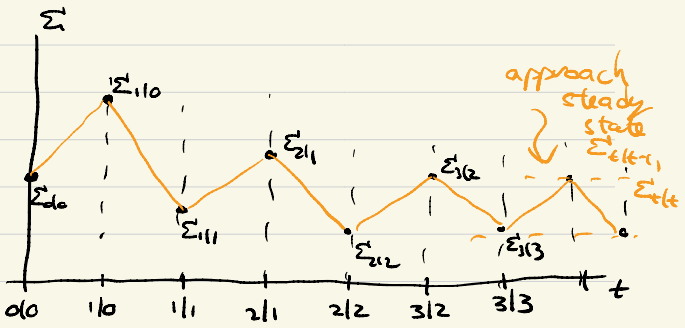
Q: How do we find Σ_{ss}?

(i) Solve the KF covariance forward in simulation until convergence (slow)

(ii) use custom algebraic solver to directly solve Riccati DARE:

Matlab: `> Σss = idare(...)` ← matrix parameters

In scalar case, the covariance executes a sawtooth trajectory:



Q: What about nonlinear dynamics and measurement models?

Extended Kalman Filter (EKF)

consider nonlinear ss model:

$$X_t = f(X_{t-1}, u_{t-1}) + W_{t-1}$$

$$W_{t-1} \sim GWN(0, Q_t)$$

$$Y_t = g(X_t) + V_t$$

$$V_t \sim GWN(0, R_t)$$

$f(\cdot), g(\cdot)$ nonlinear functions

so $(X_{0:t}, Y_{1:t})$ non-Gaussian

The EKF is an approximate Bayesian filter to give

$$\mu_{t|t} \approx E[X_t | Y_{1:t}]$$

$$\Sigma_{t|t} \approx \text{Cov}(X_t | Y_{1:t})$$

EKF Algo

Init: $X_0 \sim \mathcal{N}(\mu_{0|0}, \Sigma_{0|0})$

predict: ← nonlinear dynamics

$$\mu_{t|t-1} = f(\mu_{t-1|t-1}, u_{t-1})$$

$$\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1} \quad \text{(same as KF)}$$

update: ← nonlinear measurement model

$$\mu_{t|t} = \mu_{t|t-1} + K_t(y_t - g(\mu_{t|t-1}))$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1} \quad \text{(same as KF)}$$

where $K_t = \Sigma_{t|t-1}C_t^T(C\Sigma_{t|t-1}C_t^T + R_t)^{-1}$

where A_{t-1}, C_t are Jacobians by linearizing $f(\cdot)$ and $g(\cdot)$ about $\mu_{t-1|t-1}$ and $\mu_{t|t-1}$, respectively.

$$A_{t-1} = \left.\frac{\partial f(x_{t-1}, u_{t-1})}{\partial x_{t-1}}\right|_{\substack{x_{t-1}=\mu_{t-1|t-1} \\ u_{t-1}=u_{t-1}}}$$

(vector index)

$$= \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \in \mathbb{R}^{n\times n} \Bigg|_{\substack{x_{t-1}=\mu_{t-1|t-1} \\ u_{t-1}=u_{t-1}}}$$

$$C_t = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial g_m}{\partial x_1} & \cdots & \frac{\partial g_m}{\partial x_n} \end{bmatrix} \in \mathbb{R}^{m\times n} \Bigg|_{x_t=\mu_{t|t-1}}$$